

CSE 114 - 2026S Week 3 Prelab (Loops)

Goal

Prepare for Week 3 labwork on:

- `while` and `do-while`
- loop control variables (initialize, condition, update)
- sentinel-based input
- basic nested loops

Submission Rule

Write solutions in a single file: `week3_prelab.c`

Use separate functions and call each function inside one `main()`.

Suggested order:

```
int main(void) {
    task1_count_up();
    task2_sum_1_to_n();
    task3_even_between();
    task4_grade_average_sentinel();
    task5_nested_pattern();
    return 0;
}
```

Task 1 - Count Up (`while`)

Write `task1_count_up()`:

1. Read `n`
2. Print numbers from `1` to `n` on one line

Learning note: use one loop variable and update it every iteration.

Task 2 - Sum from 1 to n

Write `task2_sum_1_to_n()`:

1. Read `n`
2. Compute `1 + 2 + ... + n` with a `while`
3. Print the result

Learning note: accumulate into `sum` and initialize `sum = 0`.

Task 3 - Even Numbers in Range

Write `task3_even_between()`:

1. Read `a` and `b`
2. Print all even numbers in `[a, b]`

Learning note: if input order may be reversed, fix order first or handle both cases.

Task 4 - Average with Sentinel (`do-while`)

Write `task4_grade_average_sentinel()`:

1. Read grades until user enters `-1`
2. Print average of valid grades
3. If no grade entered, print `No grades`

Learning note: sentinel value should not be included in sum/count.

Task 5 - Nested Loop Pattern

Write `task5_nested_pattern()` to print:

```
1
1 2
1 2 3
1 2 3 4
```

Learning note: use outer loop for rows, inner loop for numbers in each row.

Quick Self-Check

1. Did every loop variable get initialized?
2. Does every loop update its variable?
3. Is sentinel excluded from computation?
4. Did you test with edge values (e.g., `n=0`, no grades)?